

Ans: B

Q30

$$\Delta p = m\Delta v = (8.0 \times 10^{-16})(0.2 \times 10^{-6})$$

$$= 1.6 \times 10^{-22} \text{ kg m s}^{-1}$$

$$\text{HUP: } \Delta p \Delta x \gtrsim h$$

$$\text{So } \Delta x \gtrsim \frac{6.63 \times 10^{-34}}{1.6 \times 10^{-22}} = 4.1 \times 10^{-12} \text{ m}$$

$$\approx 4 \times 10^{-12} \text{ m}$$

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